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Section A: Data Fundamentals.

1. A database is a digital repository or space, where collections of data are stored, managed & secured.

Data is essential because the modern world now has a lot of reliance on apps. ~~the~~ These apps store information on database.

Two examples for this would be social media & streaming services, & and healthcare.

For social media & email services, the profiles, contacts, messages and photos are stored and accessed ~~the~~ from the database.

Healthcare, hospitals used databases to store ~~in~~ patient information like prescriptions, patient histories & test results and track them across different departments & doctors.

2. A datawarehouse is a centralised system, that stores data from different database & apps. It collects, cleans and ~~b~~ combines that ~~or~~ information.

A company would need a datawarehouse if, for example it was a social media app. It would store user ~~info~~ information in one database & employee information in another.

3. A data lake is a centralized space used to store large volumes of data. It accommodates ~~as~~ all types of data.

- A data lake stores data like business applications, social media or streaming.
- A data warehouse stores transformed data with the intention of being used for a specified purpose.

4. A data mart is a smaller, curated & focused part of a ~~data warehouse~~ database. It has ^{of} set of tables designed to meet the needs of a specific team, line of business or community. For example the marketing department, whereas a data warehouse holds data pertaining to the whole company.

5. Structured data is highly organized information that is predefined, an example would be customer databases.

Semi-structured data is sit between structured & unstructured. It is more flexible but still retains structure. For example a json file.

Unstructured data is raw & unorganised. It is not predefined. Examples would be ~~audio~~ files, media or text.

6. Data modelling is the designing and ~~org~~ organizing of data structures. ~~It is a~~

It is important because it ^{makes} easier to ~~manages~~ manage & design database since the data is represented in a structured way.

Conceptual Data Model

It is a model that focuses on the landscape of data, by identifying the type of data, its attributes & the relationships between them.

Logical

Logical
This model focus on how the data is ~~stored~~
organized.

7. Data mining is the ~~the~~ ~~of~~ a uncovering of patterns, & valuable information from large data sets, with the use of statistical analysis & machine learning.

Risk Assessment for cybersecurity department:
Focus on target markets, by looking at the relationship ~~at~~ ^{to} behaviour of customers on certain products.

8. A specialised software that acts as the interface between database & an end-user.

a. → MySQL

↳ Open Source, organises data into structured table with columns & rows

↳ E-commerce platforms user profile management.

~~b. → PostgreSQL~~

b. → PostgreSQL

↳ Supports complex, custom data types.

↳ Healthcare systems & banking apps.

c. → Oracle Database.

↳ ~~Proprietary~~ Proprietary. Licence based used to manage massive data volumes

↳ Global supply chains.

d. MongoDB

↳ Does not use traditional tables, allows developers to change data structure

↳ For applications where data comes in various forms

9. A process that gathers data that is raw, from various systems, ~~it~~ cleans it organises it & saves it

Extract $\rightarrow$ Raw data is pulled from sources

Transform $\rightarrow$ Data is cleaned & reformatted.

Load $\rightarrow$ The data is written into a system, like a datawarehouse for the process of business analytics.

10

CSV $\rightarrow$ Comma Separated Values

$\rightarrow$ Stores data in tables made up of rows & columns. It can be opened ~~with~~ in program like ~~Excel~~ Excel.

- $\checkmark$ I would use it when I have a simple structured ~~data~~ data that needs to be shared.

JSON $\rightarrow$ JavaScript Object Notation

$\rightarrow$ This format stores data using objects and key-value pairs. Commonly used when applications & api's need to send data to each other.

- $\checkmark$ I would use JSON when working websites & applications.

Parquet -> ~~stores data~~

-> Column-based storage format designed for large datasets & analytics.

-> I would use it when working with large datasets in data analytics or big data systems.

Avro -> This is a data-serialization format, that stores data together with its schema.

-> I would use it for when exchanging data between systems.

Section B: Artificial intelligence & Machine learning.

11. Artificial intelligence is technology that allows machines & computers to ~~do~~ perform tasks that would require human intelligence.

-> Narrow AI -> ~~is~~ designed to perform a specific and limited group of tasks.

An example would be Siri for Apple. Siri can ~~per~~ perform simple tasks like make a call or find a contacts

-> General AI -> This one is designed to learn and ~~per~~ perform different & intellectual tasks like a human would.

-> An example does not exist as it's still theoretical.

12. Machine learning is when computers learn patterns from data and use these patterns to make predictions

In traditional programming, a programmer gives a computer specific rules & instructions & it follows them. ~~At~~

Machine learning is when a computer ~~is~~ is given data, learns patterns from that data, instead of every rule being directly programmed.

Supervised learning

↳ uses labelled data, this means the computer is given examples where the correct answer is already known.

↳ An example would be a model trained with emails ~~that~~ worked as "spam"

↳ Unsupervised learning there are no ~~labels~~, labelled answers. The computer tries to find the patterns of data itself.
-> A company would use it to group customers based on their buying behaviour.

Reinforcement learning

↳ learns through actions & feedback. It receives rewards when correct, or for good decisions and negative feedback for bad decisions. ~~This~~ This can be used when training systems to play games.

13. Deep learning is a type of machine learning that uses neural networks with many layers to learn patterns from large amounts of data.

AI is the bigger field, Machine learning is one way of creating AI, Deep learning is a more advanced type of machine learning.

How:

Neural networks are computer models inspired by how the ~~neural~~ neurons are connected in the human brain.

They have layers of nodes. The ~~at~~ first receives the information. The middle layer processes the patterns in the information. The final layer produces an output.

14. Natural Language Processing, is a part of AI that helps computers to understand, process and generate human language.

1. Voice assistants - such as Siri use NLP to understand questions and commands.
2. Search engines use NLP to understand what a person is searching for.
3. Chatbots use NLP to understand questions and respond to users.

15. Large Language ~~mo~~ Model is a model trained on very large amounts of text & other data, so that it can generate human-like language.

Machine learning is designed for specific tasks, like predicting sales. LLMs are designed to work with language and perform language based ~~typ~~ tasks.

Chat GPT & Claude. They can be used for research, generating code, and explaining concepts among other things.

16. Generative AI is a type of AI that generates content based on instructions or prompts.

At high levels, AI learns patterns from large amounts of ~~the~~ training data. When a user gives it a prompt, it ~~learns~~ uses what it learnt to generate a new response.

Chat GPT → generates text, analyse information and coding
Claude → generate & analyse text. and also work with documents.

Adobe Firefly → can generate & edit ~~the~~ images and other creative content.

17. A training dataset is a collection of information used to teach a machine learning model how to recognise patterns, make predictions & generate results.

It depends on the the information its given. If the data is accurate & relevant, the model has a better chance of producing useful results.

If a dataset does not properly represent certain group of people, an AI system can perform better for other people.

~~18.~~

18 Computer ~~vis~~ vision of AI that allows the computer to process and understand information from images & videos.

- Machines process pixels & patterns in an image & use the models to identify things like objects, faces, or movement.

Two real world cases.

1. → Medical imaging - x-rays can be analysed using ~~comp~~ computer vision

2. Facial recognition → computer can identify a person's face

section C: APIs, Tools & Developer concepts

19. API allows different ~~software~~ software programs to communicate to each other

A simple analogy, an order is made to the waiter at a restaurant, the waiter takes the order to the kitchen and then brings back the food to me. Like an api, the ~~request~~ request is made/sent to an api, the api communicates with another system, & it returns the result.

API are important because they help applications communicate with each other, without knowing how the other one works.

20. API is a credential used to authenticate an application when it works with an API.

They should be kept private and not written into code or ~~put~~ public repositories like Github.

* They should be ~~store~~ stored using secure methods such as secret-management tools

If an API key is exposed this could result in unauthorized ~~app~~ access.

21. REST stands for Representational State Transfer.
A REST API, is an api that follows REST principles.
It normally uses HTTP requests to communicate.

GET → used to retrieve & read information
POST → used to send information or create resource
PUT → used to update or replace existing resource
DELETE → used to ~~more~~ remove resource.

22. JSON, stands for JavaScript Object Notation, text based format used to represent & exchange structured data.

It is relatively used by humans and processed by different programming languages

Ex Example

</> JSON

```
{  
  "name": "John"  
  "age": 25,  
  "city": "Johannesburg."  
}
```

23 Prompt engineering is the process of creating clear and effective instructions for an AI system for it to produce a more useful response

It is ~~important~~ important because AI tools ~~important~~ respond better when they understand exactly what the user wants.

~~Example 1~~

Example 1

Poor ~~Data~~ Prompt

"Explain Data

Improved

Explain what data analysis is to a beginner, use simple language & give two real world examples

Example

Poor Prompt

"Make me a website.

Improved prompt:

"Create a simple responsive website for a fictional South African Coffee brand, include a ~~her~~ hero section, clear modern design. Add a products, about and contact section.

24. It is a function used to go through collection of items and apply the same operation to each item

If I have a list of numbers I want to multiply each number by two

</> JavaScript

```
let numbers = [1, 2, 3, 4]
```

```
let doubled = numbers.map(number => number * 2)!
```

```
console.log(doubled)
```

25 Cloud computing means using computing resources such as servers, storage & software over a network, usually the internet

IaaS (Amazon EC2)

↳ Infrastructure as a Service

↳ Provides computing infrastructure such as servers & networks

PaaS (Google App Engine)

Platform as a service

Provides platform where developers can build & run applications without managing infrastructure

SaaS (Google Workspace)

↳ Software as a Service

↳ Provides ready to use software through the internet.

26. Version control is a system that keeps track of changes overtime. It makes possible to see previous version

Git is a popular version control system. It is useful because it allows people to make changes to their code

Commit is saving a specific version

Branch - separate version that can be worked on without changing the original.

27. A jupyter notebook is an interactive environment where one can write, run code, & view results & add explanation to them.

commonly used with Python

They allow code, data, charts & explanations to be saved together, makes it easier to experiment with data.

28. Python is a general-purpose programming language used for software that makes it easier to work with data.

Pandas $\rightarrow$ Analysing structured data, like tables.

Numpy $\rightarrow$ Numerical calculations

Matplotlib $\rightarrow$ Create graphs and visualization.

Scikit-learn $\rightarrow$ Machine learning

29. Data Privacy is about protecting people's personal ~~and~~ information that is collected, stored and ~~secretly~~ used responsibly.

The information is sensitive, people should have some control over how their information is collected & used.

GDPR is a data protection law.

It applies to countries in the EU.

30. Data ethics using the data collected in an appropriate or safe way.

Using data Using personal data without permission
- The data could be used for on unknown reasons, to the people.

2. selling personal data without proper consent.
violates people's trust

using biased AI
• certain groups could be mistreated.

31. The process of creating rules & policies on the organisations of data.

Data quality → make sure its accurate

Data security → Protection from unauthorized access

Data Ownership → Responsibility of specific data.

Policies & standards → Rules for data collection

Data Access → The right people have access

Compliance → Practices must follow relevant laws.

32. When AI generates results that unfairly favours or disadvantage certain groups of people.

~~Is an app~~

If the system is trained on old data, that reflected gender imbalances, the patterns learned could disadvantage women.

Qualified people could be judged by characteristics that should not influence the decision.

Role	Responsibility	Skills	Tools
Data Analyst	Analyses existing data.	SQL, Excel stats, visualisation	Excel, SQL
Data Scientist	Uses stats and Machine learning to make prediction	Python, Stats ML, SQL	But Python Jupyter
Data Engineer	Builds & maintains Systems	Programming, databases	Spark, Cloud platforms
AI Engineer	Builds & maintains Machine learning systems	Python, ML, AI, APIs software development.	Python Tensorflow PyTorch.

34. BI, is used by businesses, by the use of data to make well ~~informed~~ informed decision

35 Machine learning and deep search. Honestly had ^{been} not anticipated it would be with the data ~~conv~~ conversation.

36, More AI will collect data, we need to be able to work with that.

37 APIs, google